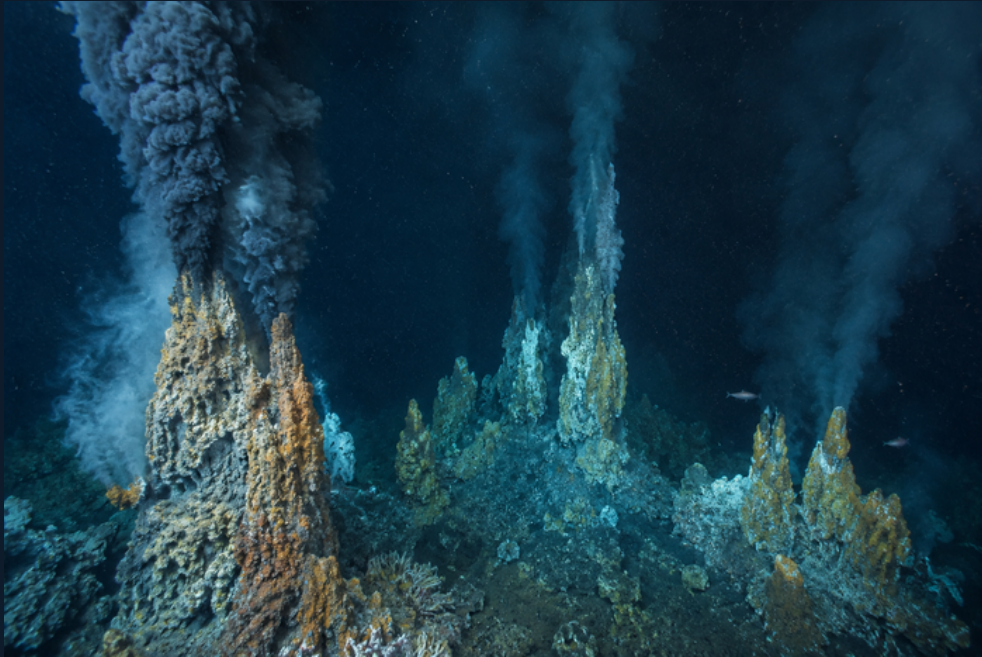


HYDROVENT_FIELD

USER MANUAL



Procedural hydrothermal vent field generator & prebiotic-chemistry simulator

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1. Introduction

hydrovent_field is a procedural generator of mid-ocean-ridge hydrothermal vent fields and a physically grounded simulator of prebiotic-molecule concentration near those vents. It was built to test literature-cited hypotheses — not to illustrate them — so every physical model in the software is either validated against published field data or explicitly flagged as an untested, exploratory hypothesis where no such data exists.

The software combines:

- A fractal (diamond-square) terrain generator that lays a realistic mid-ocean-ridge axial valley and randomly clustered vents (black smokers, white smokers, diffuse-flow vents) on top of it;
- A buoyant-plume physics model (Morton-Taylor-Turner) that computes how each vent's fluid dilutes and rises through the stratified water column;
- Four literature-cited concentration mechanisms for prebiotic molecules (plume dilution, mineral-pore thermophoresis, mineral-surface adsorption, and proton-gradient compartmentalization), plus an original, clearly-labelled exploratory acoustic-concentration hypothesis;
- Single-run, ensemble (up to 10,000 runs) and nested variance-decomposition execution modes, with optional parallel execution across CPU cores;
- A statistics/report generator that turns an ensemble into a self-contained HTML report with descriptive statistics, charts, and (when applicable) sensitivity/variance tables.

This manual covers the desktop graphical interface (`python gui.py`), every option it exposes, and every figure/file a run produces. It does not cover the author's personal manuscript-preparation scripts, which ship in the working copy but are excluded from the public distribution and from the GUI itself.

2. Installation

2.1 Requirements

Python **3.10 or newer** (tested on 3.10–3.12). The GUI uses Tkinter, part of the Python standard library — already bundled with the official Windows and macOS installers from python.org. On Linux it is often a separate system package.

On Debian/Ubuntu, if needed: `sudo apt install python3-tk`

Package	Version	Used for
numpy	≥ 1.24	Numerical arrays, RNG (seeded reproducibility)
scipy	≥ 1.10	Latin Hypercube sampling (sensitivity sweeps), ODE/statistics
matplotlib	≥ 3.11	All 2D/3D figures and the embedded ensemble charts
pillow	≥ 10.0	Splash image, PNG loading/zoom in the image viewer
pytest	≥ 7.0	Running the automated test suite (optional, for verification)
pyvista (optional)	≥ 0.48	Only for the "Artistic view" 3D render (checkbox in 3D visualization)

2.2 Step-by-step install

The project is not a pip-installable package — there is no `setup.py` or `pyproject.toml`. It is run directly from a local copy of the repository:

```
git clone https://github.com/cesarrrlamara1/hydrovent_field.git

cd hydrovent_field

python -m venv .venv

.venv\Scripts\activate      (Windows)

source .venv/bin/activate   (macOS / Linux)

pip install -r requirements.txt
```

The virtual environment (venv) step is optional but recommended, to keep these package versions isolated from other Python projects on your machine.

2.3 Launching the application

From the project folder, with the environment activated:

```
python gui.py
```

This opens the welcome screen described in Chapter 4. An equivalent command-line interface (`python fumarola_field.py --seed 42 --runs 1` etc.) also exists for scripting/batch use, but is outside the scope of this manual.

2.4 Verifying the install (optional)

To confirm everything is working, run the automated test suite:

```
pytest tests/
```

2.5 Where output files go

By default every run is written under an `outputs/` folder created inside the project directory, regardless of where you launched the app from. This can be changed per-run in the GUI's Output section (§ 4.9).

3. Quick Start

A minimal first run, using every default value except the seed:

- Run `python gui.py`. The welcome screen appears (Fig. 3.1).
- Pick **English** or **Português** at the top, then click **Continue**. This choice fixes the language for the whole session.
- The main window opens (Fig. 3.2) with the default parameters already filled in and “Single run” selected.
- Optionally type a number into **Base seed** (e.g. 42) so the result is exactly reproducible later.
- Click **Run**. Progress is printed live in the **Log** box at the bottom.
- When it finishes, the 2D field map appears automatically in the **Images** tab (Fig. 3.3); use the **Image** dropdown to browse the other figures for that run.

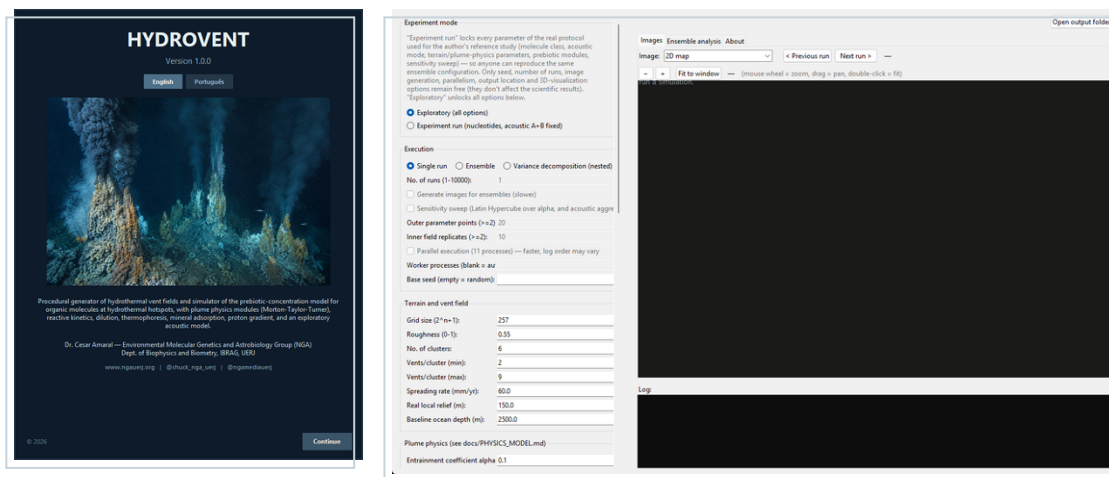


Fig. 3.1 (left) — welcome screen. Fig. 3.2 (right) — main window, default state.

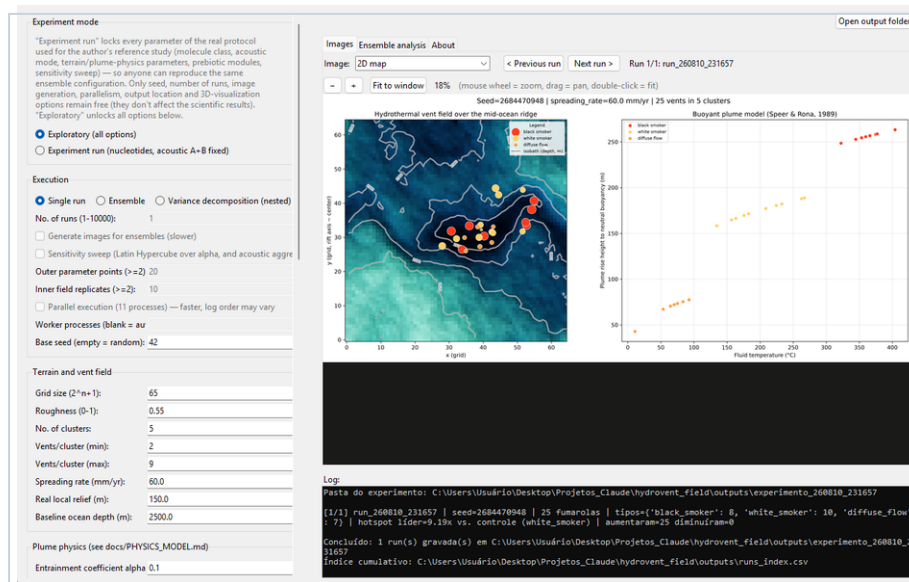


Fig. 3.3 — Images tab after a finished run, showing the 2D field map.

4. The Graphical Interface

This chapter walks through every screen, tab and control, in the order they appear on screen. Each option is described in one or two lines; see the on-screen tooltips/labels (and, for the physics parameters, docs/PHYSICS_MODEL.md) for full scientific detail.

4.1 Welcome screen

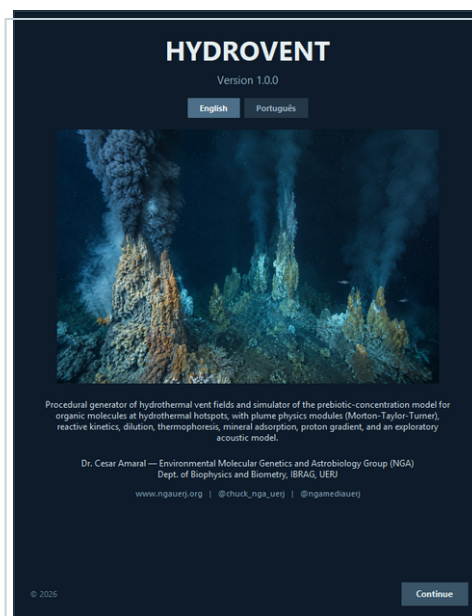


Fig. 4.1 — welcome/splash screen.

- **English / Português** — language selector; English is selected by default. The choice applies to the entire main-window session (there is no way to switch language later without restarting the app).
- Hero photograph, software name/version and short description/credits — purely informational.
- **Continue / Continuar** — closes the welcome screen and opens the main window.

4.2 Main window layout

The main window is split into a scrollable parameter form on the left, and a results panel on the right (image viewer / ensemble statistics / about, plus a live log console at the bottom). A **File** menu (Open existing experiment..., Resume aborted experiment..., Exit) and a **Help** menu (About) sit at the very top.

4.3 Experiment mode

Option	What it does
Exploratory (all options)	Default. Every option below is editable.
Experiment run (nucleotides, acoustic A+B fixed)	Locks terrain, physics, molecule class, acoustic mode and prebiotic modules to the exact protocol used in the author's reference study, so anyone can reproduce the same configuration. Only seed, run count, image generation, parallelism and output location stay free. Also switches Execution to Ensemble and suggests 1000 runs.

4.4 Execution

Option	What it does
Single run	Default. Runs the simulation once.
Ensemble	Runs the simulation N times with different, derived seeds — for statistics over many random vent fields.
Variance decomposition (nested)	Alternate mode: N_{outer} sampled parameter points $\times$ N_{inner} field replicates each, to split result variance into a “parametric uncertainty” part and a “random vent placement” part.
No. of runs (1–10000)	Ensemble size. Default 1 (only editable in Ensemble mode).
Generate images for ensembles (slower)	Off by default — ensembles normally save only data (JSON/CSV/metadata), which is much faster.
Sensitivity sweep	Off by default. Draws entrainment α (and, if an acoustic mode is active, particle radius/density) from a joint Latin-Hypercube design across the ensemble, over documented uncertainty ranges.
Outer parameter points (≥ 2)	N_{outer} for variance decomposition. Default 20.
Inner field replicates (≥ 2)	N_{inner} for variance decomposition. Default 10.
Parallel execution (N processes)	Off by default. Distributes ensemble/vardecomp runs across CPU cores (one process per core minus one). Same seeds/results as sequential, log order may vary.
Worker processes (blank = auto)	Overrides the automatic worker count used by parallel execution.
Base seed (empty = random)	Root RNG seed. Every run in an ensemble derives its own seed from this one (via SeedSequence), so the same base seed always reproduces the same field(s). Always editable, even in “Experiment run” mode.

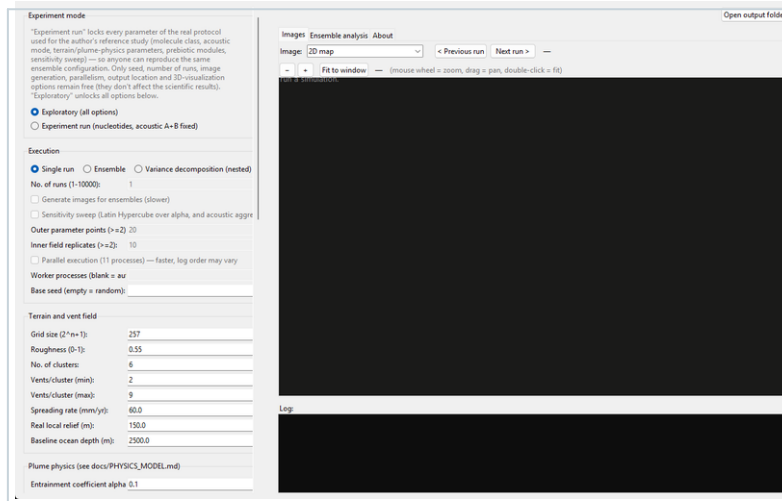


Fig. 4.2 — Experiment mode + Execution, top of the form.

4.5 Terrain and vent field

Option	What it does
Grid size (2^n+1)	Terrain resolution, e.g. 65/129/257/513. Default 257 — larger is slower but more detailed.
Roughness (0–1)	Fractal roughness of the terrain. Default 0.55.
No. of clusters	Vent clusters along the ridge axis. Default 6.
Vents/cluster (min / max)	Range of vents drawn per cluster. Default 2–9.

Option	What it does
Spreading rate (mm/yr)	Ridge spreading rate — slow <40, intermediate 40–90, fast >90. Default 60.
Real local relief (m)	Real terrain relief variation of the surveyed area, typically 50–300 m. Default 150.
Baseline ocean depth (m)	Ambient depth at the ridge crest, typically 2000–3000 m. Default 2500.

4.6 Plume physics

Option	What it does
Entrainment coefficient α	Morton-Taylor-Turner plume model parameter. Default 0.1; measured field range 0.07–0.18 (Rona et al. 2006).
Brunt-Väisälä frequency N (s^{-1})	Stratification of the water column. Default $7.9 \times 10^{-4} s^{-1}$, a verified oceanic ridge value.
Basin (Fe kinetics)	Atlantic (TAG) or Pacific (EPR) — sets the Fe(II) oxidation half-life, which differs between basins (Field & Sherrell, 2000).
Export full plume profiles (slower)	Off by default. Writes the full vertical profile (dilution, temperature, concentration) of every vent's plume to a CSV file.

The screenshot shows a software configuration window with several sections:

- Terrain parameters:**
 - Spreading rate (mm/yr): 60.0
 - Real local relief (m): 150.0
 - Baseline ocean depth (m): 2500.0
- Plume physics (see docs/PHYSICS_MODEL.md):**
 - Entrainment coefficient alpha: 0.1
 - Brunt-Väisälä frequency N (s^{-1}): 0.00079
 - Basin (Fe kinetics): atlantic
 - ☐ Export full plume profiles (slower)
- 3D visualization:**
 - ☒ Generate stylized 3D scene
 - ☐ Also generate true-scale scene
 - ☐ Also generate artistic view (non-scientific, requires pyvista)
 - Vertical exaggeration (z_{exag}): 25.0
 - Chimney scale: 1.0
 - Camera - elevation: 55.0
 - Camera - azimuth: -50.0
 - Real domain extent (m): 1200.0
- Prebiotic concentration modules:**
 - Molecule class: Amino Acids
 - Pore aspect ratio (blank = 10): 10
 - ☒ Dilution / plume advection
 - ☒ Thermophoresis in mineral pores
 - ☒ Adsorption on mineral surfaces
 - ☒ Proton gradient (alkaline compartments)
- Acoustic model (exploratory hypothesis):**
 - Tests whether the real sound field of hydrothermal vents (Crone et al. 2006) could concentrate prebiotic molecules. No experimental validation — see docs/PHYSICS_MODEL.md.
 - ☒ Off
 - ☐ A — Acoustic streaming (dissolved solute)

Fig. 4.3 — Terrain, Plume physics and (start of) Prebiotic modules sections.

4.7 3D visualization

Option	What it does
Generate stylized 3D scene	On by default. Vertically-exaggerated 3D render with chimney shapes and plume point clouds.
Also generate true-scale scene	Off by default. Same 3D scene at real 1:1:1 proportions — shows how flat a real vent field actually looks.
Also generate artistic view (non-scientific, requires pyvista)	Off by default. Photorealistic PyVista/VTK render with rock/mineral textures, for presentation only — geometry is real, lighting/materials are aesthetic choices.

Option	What it does
Vertical exaggeration (z_exag)	Relief stretch factor for the stylized scene. Default 25.
Chimney scale	Visual height multiplier for chimneys. Default 1.0.
Camera elevation / azimuth	3D camera angles. Defaults 55° / –50°.
Real domain extent (m)	Horizontal size of the simulated area in meters, used by the true-scale/artistic scenes and acoustic field plot. Default 1200.

4.8 Prebiotic concentration modules

Option	What it does
Molecule class	Amino Acids (default) / Nucleotides & Nitrogenous Bases / Lipids (Amphiphilic Fatty Acids) / Sugars (Glyceraldehyde, Ribose). Sets the baseline synthesis rate, thermal window, Soret coefficient and adsorption/proton-gradient response used below.
Pore aspect ratio (blank = 10:1)	Length/width ratio of the mineral pore used by the thermophoresis module (Nucleotides class only).
Dilution / plume advection	On by default. The baseline mechanism: concentration falls simply because the plume dilutes as it rises.
Thermophoresis in mineral pores	On by default. Soret-effect concentration inside mineral pores along a thermal gradient.
Adsorption on mineral surfaces	On by default. Concentration via binding to iron-sulfide mineral surfaces.
Proton gradient (alkaline compartments)	On by default. Concentration via proton-motive-force compartmentalization (Russell & Martin hypothesis).

4.9 Acoustic model (exploratory hypothesis)

Option	What it does
Off	Default. Acoustic module disabled entirely.
A — Acoustic streaming (dissolved solute)	Concentration of dissolved molecules by acoustic streaming around the vent's own sound field.
B — Particle trapping (Gor'kov)	Concentration of suspended particles at acoustic pressure nodes (Gor'kov potential).
A + B — Both combined	Both mechanisms together; this is the setting used by the locked “Experiment run” protocol.
Custom particle radius (μm) / density (kg/m³)	Only for modes B/A+B. Blank = use the literature-cited particle population instead of one custom particle.
Test idealized upper bound (all vents in phase, same frequency)	Off by default (incoherent summation between independent vents — the realistic case). On = optimistic best-case scenario, for testing how much that assumption matters.

The screenshot shows the 'Acoustic model (exploratory hypothesis)' and 'Output' sections of the software. The 'Acoustic model' section includes a text box for 'Real domain extent (m):' with the value '1200.0'. Below it, the 'Prebiotic concentration modules' section has a 'Molecule class:' dropdown set to 'Amino Acids' and a 'Pore aspect ratio (blank = 10):' text box. Four checkboxes are checked: 'Dilution / plume advection', 'Thermophoresis in mineral pores', 'Adsorption on mineral surfaces', and 'Proton gradient (alkaline compartments)'. The 'Acoustic model (exploratory hypothesis)' section contains a paragraph of text, a radio button selected for 'Off', and three other radio buttons for 'A — Acoustic streaming (dissolved solute)', 'B — Particle trapping (Gor'kov)', and 'A + B — Both combined'. Below these are two text boxes for 'Custom particle radius, μm (b)' and 'Custom particle density, kg/m'. A paragraph of text follows, and a checkbox for 'Test idealized upper bound (all vents in phase, same frequency)' is unchecked. The 'Output' section has a text box for the output path set to 'C:\Users\Usuário\Desktop\Projetos_Claude\hydrovent_field\outputs' and a 'Base filename:' text box set to 'fumarola_field'. At the bottom is a 'Run' button and a 'Ready.' status indicator.

Fig. 4.4 — Acoustic model and Output sections (bottom of the form).

4.10 Output & Run

Option	What it does
Output folder (+ “...” browse button)	Root folder where each run's subfolder is created. Default: out puts / inside the project.
Base filename	Filename prefix used for every file generated by a run. Default fumarola_field.
Run	Starts the simulation in a background thread (the window stays responsive); disabled while a run is in progress.

4.11 File and Help menus

- **File** → **Open existing experiment...** — load the images/statistics of a previously completed experiment folder, as if it had just finished running.
- **File** → **Resume aborted experiment...** — lists incomplete experiment folders (interrupted by a crash or manual stop) and completes only the missing runs, with the exact same derived seeds as if it had never stopped.
- **File** → **Exit / Help** → **About** — quit the app / jump to the About tab (§ 4.14).

4.12 “Images” tab — the result viewer

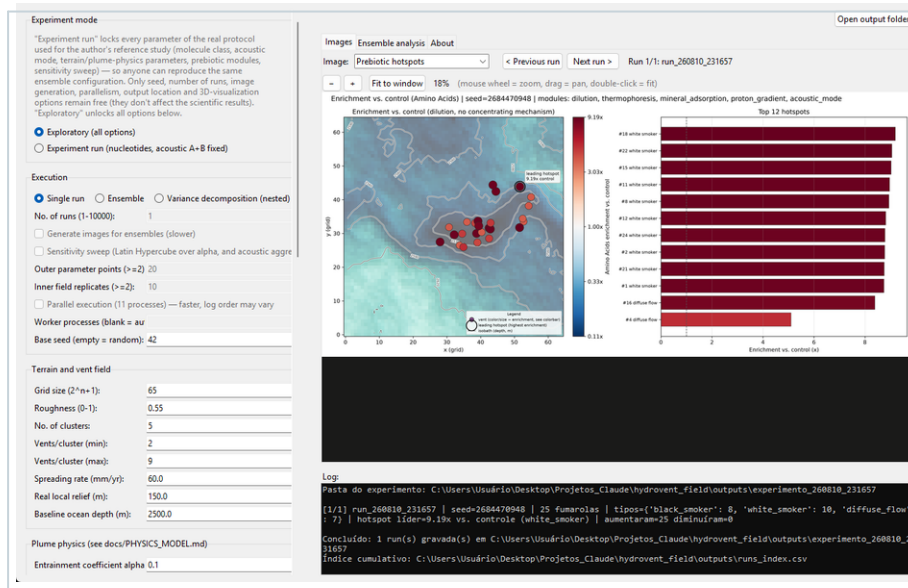


Fig. 4.5 — Images tab, showing the hotspot-enrichment map of the run in Fig. 3.3.

- **Image** dropdown — switch between the up-to-10 figure types produced by the current run (a given type may be greyed out/unavailable if that feature was off for that run).
- **< Previous run / Next run >** — page through the runs of a multi-run experiment.
- **Zoom – / +, Fit to window** — mouse wheel also zooms (centered on the cursor, 5–400%); left-click-drag pans; double-click fits the image to the window.
- **Open output folder** (top-right, always visible) — opens the current run's output folder in Windows Explorer.

4.13 “Ensemble analysis” tab

Populated automatically whenever a run produces more than one field (Ensemble or Variance decomposition modes).



Fig. 4.6 — Ensemble analysis tab after a 10-run ensemble.

- **Generate statistical report (HTML)** — writes a self-contained `ensemble_statistics_report.html` to the experiment folder and opens it automatically (see § 5.5).
- **Descriptive statistics** — leading-hotspot and pooled enrichment vs. control, vents increased/decreased/unchanged, max/mean concentration and vent count, all as `n / mean / std / min / median / max` across the runs.
- **Charts** — 4 panels: enrichment histogram, leading-hotspot histogram, leading-hotspot-type bar chart, and no.-of-vents vs. leading-hotspot scatter (see Fig. 5.8 for the full-size version).
- **Experiment runs table** — one sortable row per run (click a column header to sort): seed, vent count, leading hotspot, mean enrichment, vents increased/decreased, leading hotspot type.
- For a *Variance decomposition* run, an extra panel shows live stochastic-vs-parametric variance fractions and (if computed) Sobol' sensitivity indices per swept parameter.

4.14 “About” tab

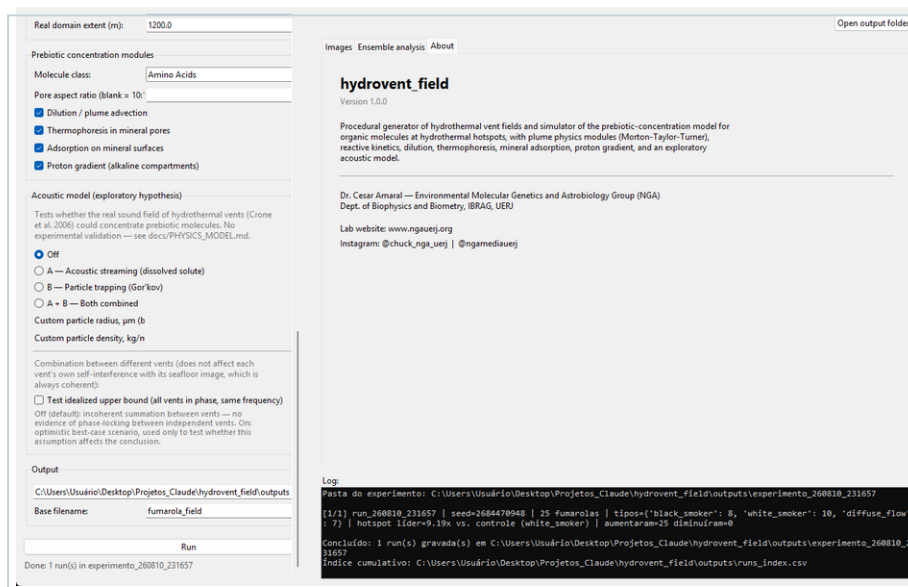


Fig. 4.7 — About tab: app name/version, description, author credits and links (same content as the welcome screen, § 4.1).

5. Understanding Your Results

5.1 Where files are saved

Every click of **Run** creates one timestamped *experiment* folder, containing one timestamped *run* subfolder per simulated field:

```
outputs/

  experimento_YYMMDD_HHMMSS/

    experiment_metadata.json

    (ensemble_statistics_report.html, if generated)

    run_YYMMDD_HHMMSS/

      fumarola_field.png / .json / .csv

      fumarola_field_hotspots.png

      metadata.json

      prebiotic_hotspots.csv

      ... (see table below)
```

A single cumulative outputs/runs_index.csv logs one summary row per run ever executed, across every experiment.

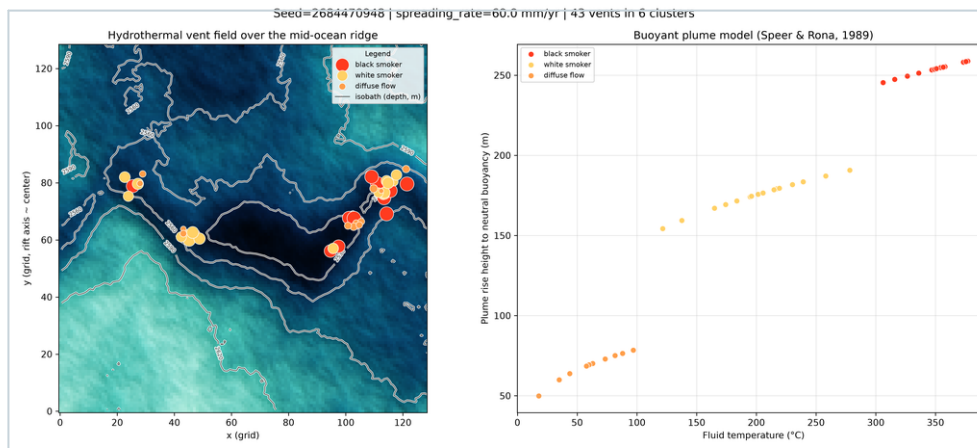
5.2 Per-run files

File	Contents
.png	2D field map — see Fig. 5.1.
_hotspots.png	Hotspot/enrichment map — see Fig. 5.2.
_3d.png	Stylized 3D scene — see Fig. 5.3.
_3d_truescale.png	True-scale 3D scene — see Fig. 5.4 (only with “true-scale” on).
_artistic.png	Artistic render — see Fig. 5.5 (only with “artistic view” on).
_acoustic.png	Acoustic enrichment field — see Fig. 5.6 (only if an acoustic mode is active).
module<name>.png	Gradient map for one concentration mechanism — see § 5.3d (one per active module).
.json / .csv	Raw per-vent field data (position, type, temperature, plume height...).
_plume_profiles.csv	Full vertical plume profile per vent (only with “export plume profiles” on).
prebiotic_hotspots.csv	One row per vent with computed concentration/enrichment for the chosen molecule class.
metadata.json	Full run summary — every parameter, every file path, prebiotic/acoustic summary statistics.

5.3 Image gallery

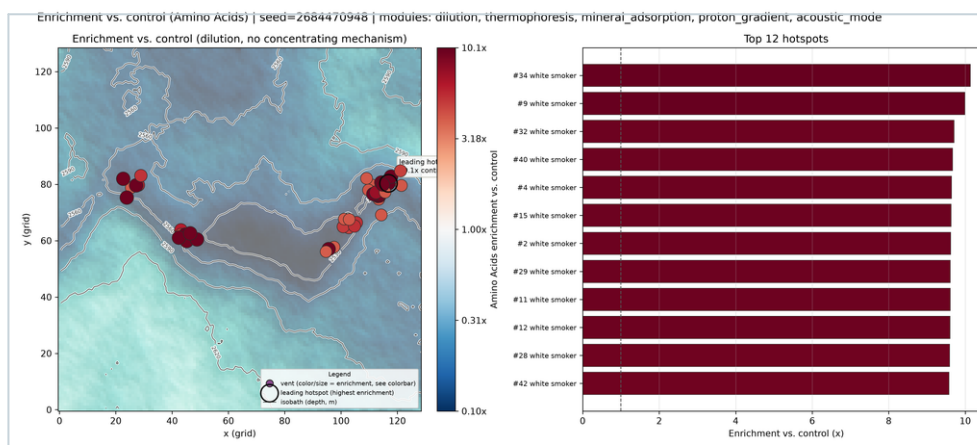
All figures below come from the same real, small sample run used throughout this manual (seed 42, 25 vents in 5 clusters) — actual software output, not mock-ups.

Fig. 5.1 — 2D field map

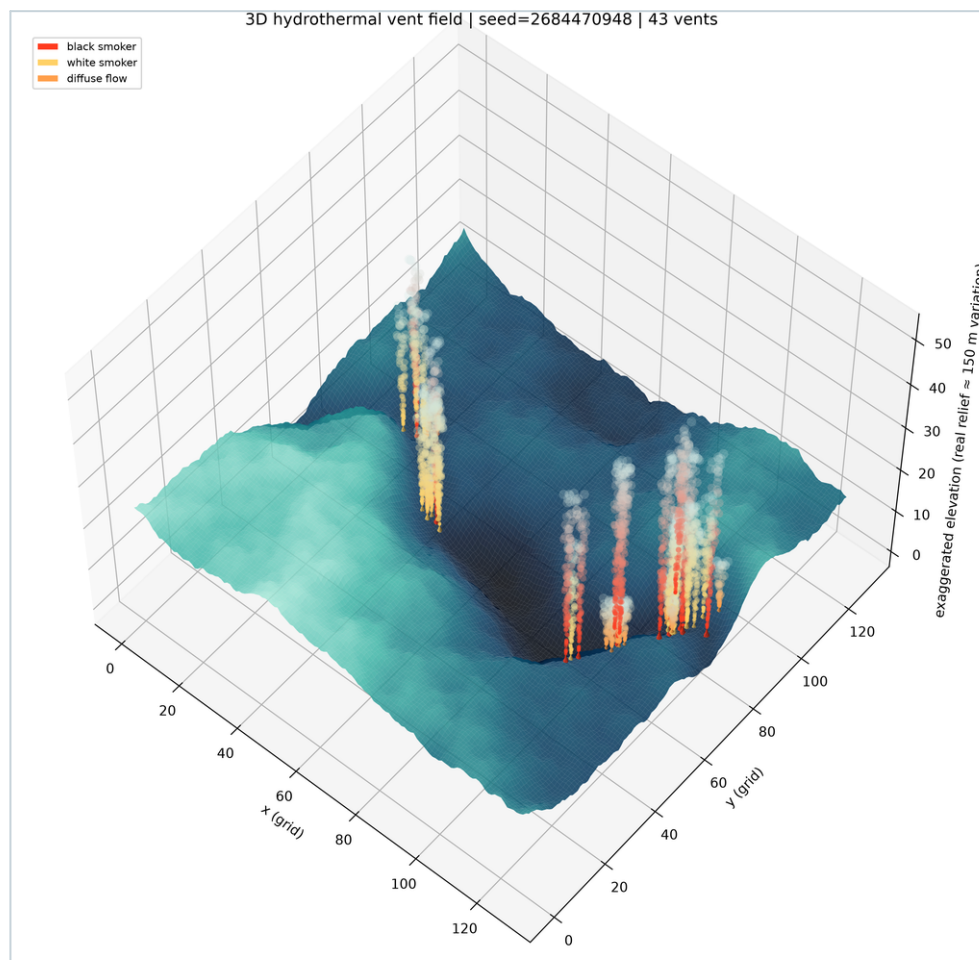


Shaded-relief top view of the seafloor with every vent placed, colored/sized by type and temperature, next to a scatter of fluid temperature vs. plume rise height. This is the software's basic “what did it generate” figure.

Fig. 5.2 — Hotspot / enrichment map

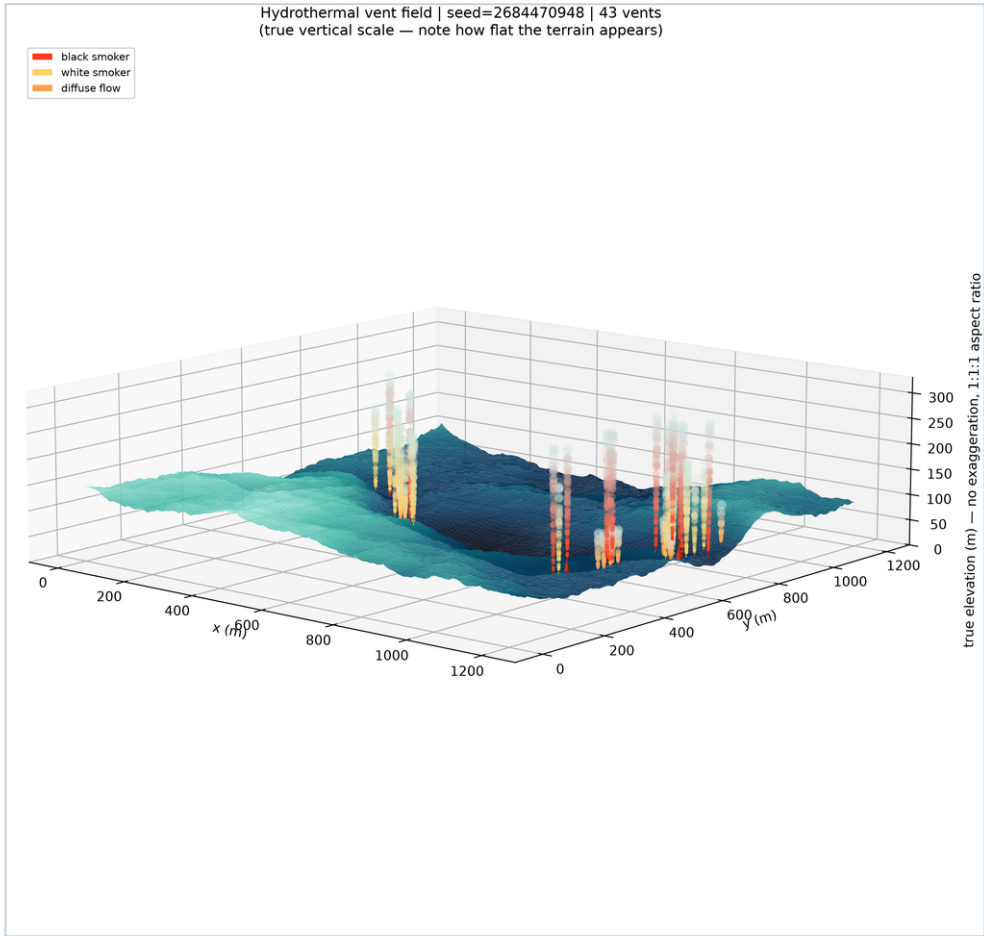


Same field, now colored by how much more concentrated the chosen prebiotic-molecule class is at each vent, relative to a no-concentration-mechanism control — plus a ranked list of the strongest hotspots. This is the figure that answers “where does the model predict prebiotic molecules build up the most?”.

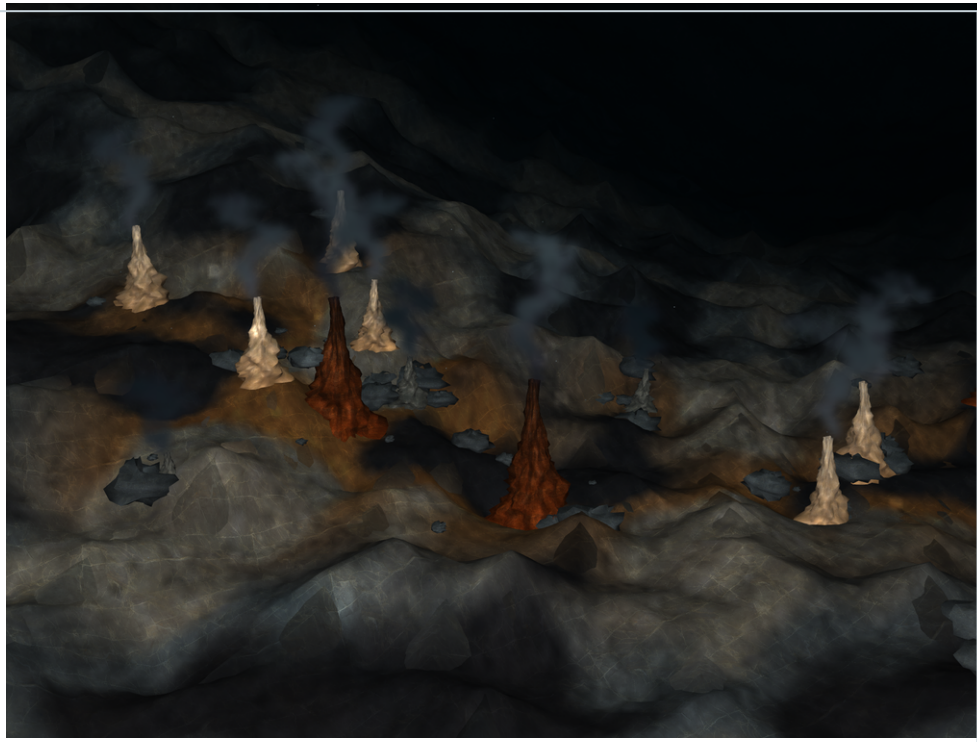
Fig. 5.3 — 3D scene (stylized)

Vertically-exaggerated 3D rendering with chimney shapes and smoke-plume point clouds — relief and plume height are stretched for visibility, not to true scale.

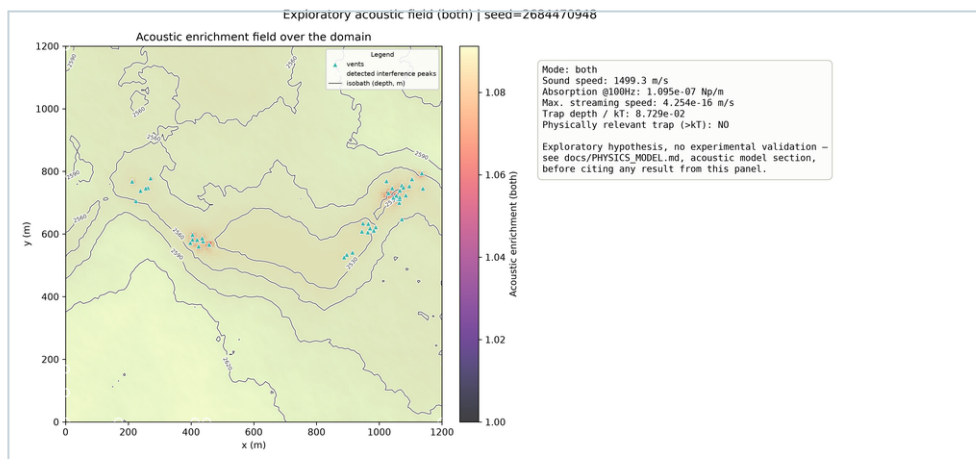
Fig. 5.4 — 3D scene (true scale)



The same 3D scene at real 1:1:1 proportions, included to show how flat and small a real vent field actually looks compared to the stylized version above.

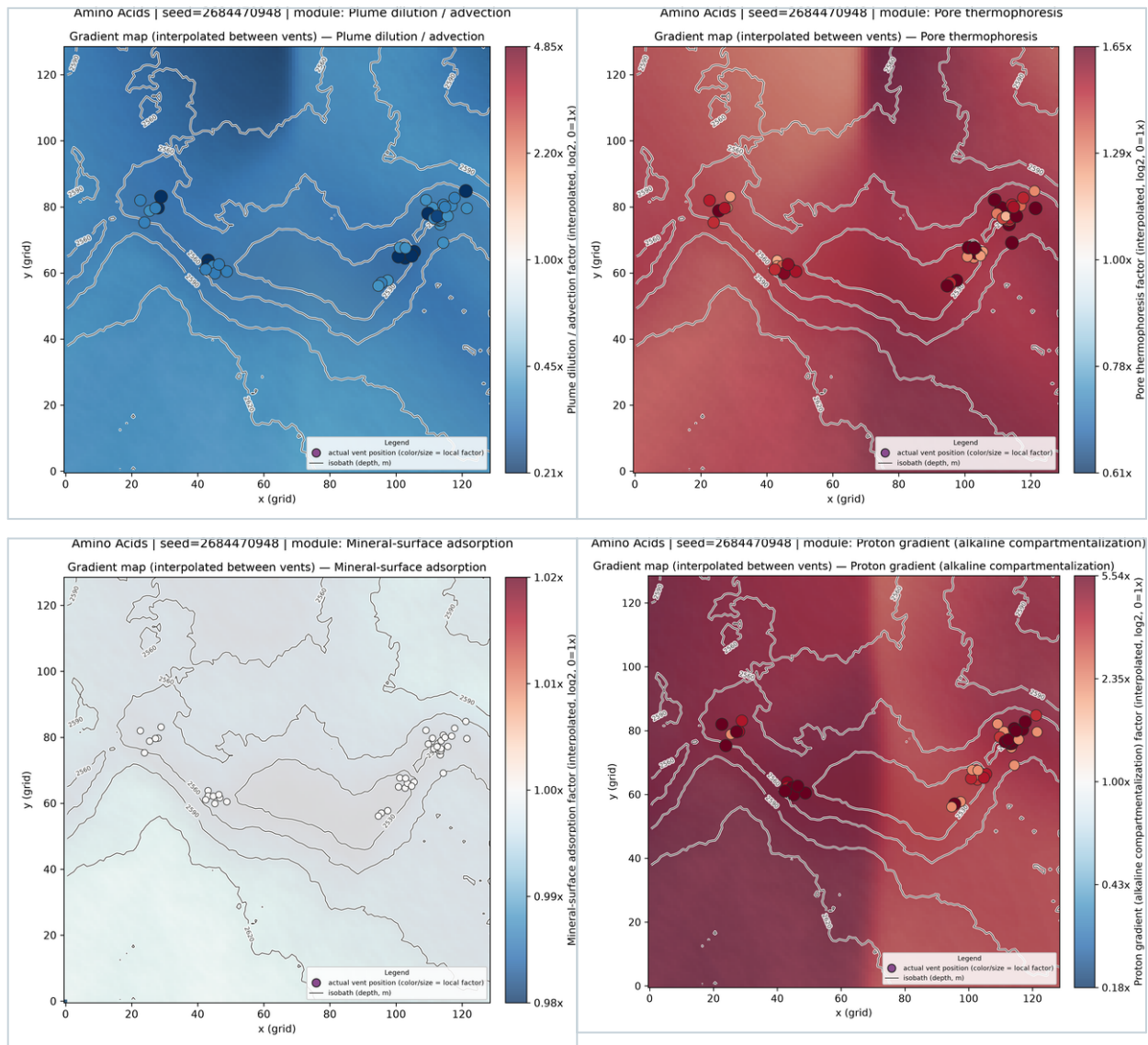
Fig. 5.5 — Artistic view (non-scientific)

Photorealistic PyVista/VTK render with rock/mineral textures and lighting calibrated against a real reference photograph. Geometry comes from the same simulated data; colors and lighting are aesthetic choices, not measurements — for presentations, not analysis.

Fig. 5.6 — Acoustic enrichment field (exploratory hypothesis)

Domain-wide heat map of predicted sound-driven concentration enrichment, plus a diagnostics panel stating whether the effect is physically plausible (comparing acoustic trap depth to thermal energy kT). On-figure labelling makes clear this is an untested, exploratory hypothesis, not a validated mechanism.

Fig. 5.7 — Concentration-mechanism gradient maps



One interpolated gradient map per active concentration module — a smoothed visualization of that single mechanism's effect between discrete vent values, not a resolved physical field. Generated for whichever modules are switched on in § 4.8 (Dilution, Thermophoresis, Mineral adsorption, Proton gradient, clockwise from top-left above).

5.4 Experiment- and ensemble-level outputs

File	Contents
experiment_metadata.json	Base seed, run count, image flag, and the full parameter set for the whole experiment — written as soon as the run starts (needed later to resume an aborted experiment).
outputs/runs_index.csv	One cumulative row per run ever executed, across every experiment — seed, vent counts, module toggles, acoustic mode, headline enrichment results. The master log for finding/comparing runs.

5.5 Statistical report (HTML)

Generated on demand from the Ensemble analysis tab (§ 4.13) as `ensemble_statistics_report.html` inside the experiment folder, and opened automatically in your browser. It is a plain data report — no discussion/interpretation text — containing: a metadata table (seed, module configuration, molecule class); a descriptive-statistics table with 95% bootstrap confidence intervals; the same 4-panel chart shown live in the GUI (Fig. 5.8); a full per-run results table; and, only if the ensemble used a sensitivity sweep or variance-decomposition mode, a driver-analysis and/or variance/Sobol' table.

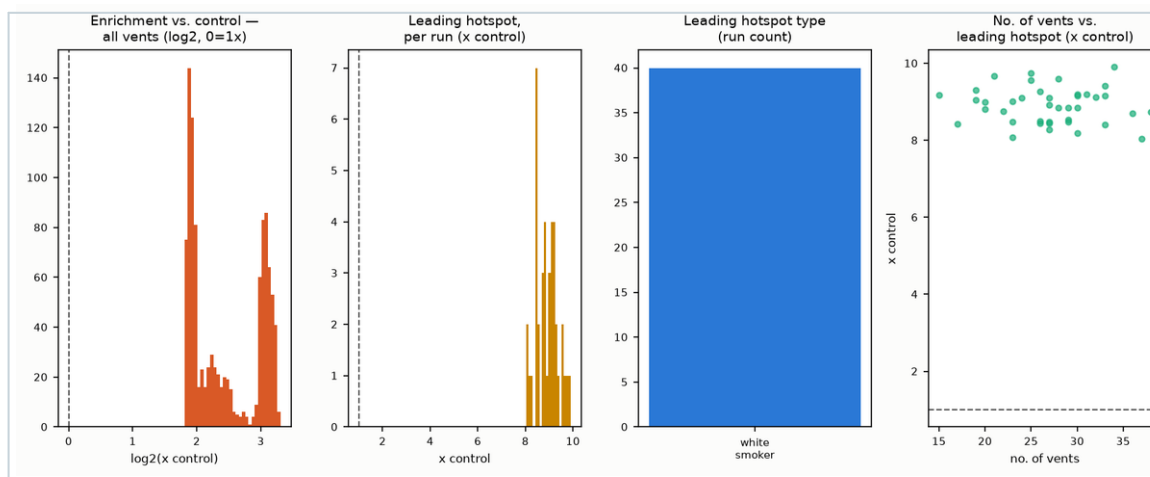


Fig. 5.8 — the 4-panel ensemble chart (log2 enrichment histogram, leading-hotspot histogram, leading-hotspot-type bar chart, vent-count vs. leading-hotspot scatter), from a 40-run sample ensemble. Embedded both in the Ensemble analysis tab and in the HTML report.

5.6 Variance-decomposition outputs

A “Variance decomposition (nested)” run additionally writes `vardecomp_runs.csv` (raw per-run rows) and `vardecomp_summary.json` (the computed stochastic-vs-parametric variance split, plus Sobol' global-sensitivity indices when available) to the experiment folder.

6. Practical Tips

- **Reproducibility** — always set a **Base seed** before a run you want to cite or compare later; leaving it blank draws a random one that is only printed in the log/metadata after the fact.
- **Start small** — use a small grid (65 or 129) and Single run to check your parameters before committing to a large grid (257+) or a large ensemble; runtime grows quickly with grid size and run count.
- **Ensemble images stay off by default** — turn on “Generate images for ensembles” only if you actually need per-run figures; otherwise an ensemble is much faster with just data files.
- **Crashed mid-ensemble?** Use **File** → **Resume aborted experiment...** instead of starting over — it completes only the missing runs, with identical seeds.
- **“Experiment run” mode** is the fastest way to exactly reproduce the author's reference protocol — only touch seed/run-count/output options while it is selected.
- Click **Open output folder** (top of the Images tab) at any time to jump straight to the current run's files in Explorer.

For the full physical/mathematical model behind every option in this manual, see docs/PHYSICS_MODEL.md (or the English version, docs/PHYSICS_MODEL_EN.md) in the project folder.